

RESPONSE TO THE R12 REFEREE REPORT

We thank the referee for the careful examination of the development snapshot and for identifying the evidence needed to assess the revised method. This revision develops the same paper, *Neural Bellman Operators*. Its central economic subject remains policy evaluation and feasible improvement in continuous-time models with recursive utility, endogenous preferences, temporal selves, and strategic interaction. We have completed the evidence account, extended the mathematical connection between evaluation and economic performance, added a finite-observation implementation theorem, and reorganized the article around these results and their numerical tests.

The report reviewed the development commit beginning 39344d49 and is preserved at the review commit beginning 65110ed2. The report accurately identifies what was absent from that development snapshot. The present numerical record comes from the completed primary calculation at source 95053e72 and the completed fixed-work extension at source c8299feb. Their original generating sources remain separately identified. The present integration binds their recovered weights, raw arrays, protocols, worker records, independently recomputed endpoints, and generated tables to the manuscript; it does not describe the recovered fits as newly trained by the publication commit.

PRINCIPAL RESULTS AND CHANGES

The completed primary study contains ninety fitted policies, with ten streams in each of dimensions ten, twenty, and fifty for NBO, direct policy optimization, and affine policies. All 180 primary schedule comparisons, covering both the origin and the declared finite population, have positive simultaneous lower improvement endpoints. All 120 direct NBO-minus-comparator intervals contain zero. These results establish improvement over the feasible analytical schedule for the recorded policies and leave their direct method rankings unresolved at the reported precision. Both conclusions appear in the main text and in every seed-level table.

The fixed-work extension contains twenty-eight fits, fifty-six policy evaluations, and forty-two direct comparisons, including one repeated environment. It compares exact iteration and simulator-visit budgets, reports common certified-gain targets,

and includes raw-costate improvement without critic regression. The independent-reference mechanism calculation finds larger critic mean squared error and larger critic-greedy Hamiltonian gaps in all six principal dimension-stream panels. The corresponding economic comparisons are reported separately. This is a measured property of the fitted evaluation block, and it is retained in the interpretation of the method.

Proposition 11 strengthens the theoretical connection: it links costate regression error, rollout bias, and feasible-action error under the candidate's discounted state occupation law to economic improvement. Proposition 14 specifies the comparison filtration. Proposition 15 supplies an explicit transfer from a protected innovation controller to a controller using discrete noisy state measurements. Eighteen completed sensing cells accompany that theorem as implementation diagnostics. A four-grid scalar reference replay reproduces twenty-eight original arrays exactly in the recorded environment and adds equation residuals, full action diagnostics, and five initial-state comparisons.

BLOCKING COMMENTS

B1. Materialize the completed revision. The article now uses the integrated root sources, with its tables materialized from the complete records. The primary calculation includes all ten prescribed fitting streams, three dimensions, three primary methods, the origin and population designs, auxiliary policies, stresses, checkpoint comparisons, and the scalar reference. Its full record contains 244 policy-evaluation rows and 120 direct comparisons; all 364 corresponding raw statistical arrays have been independently replayed. The fixed-work extension supplies the additional fifty-six evaluations and forty-two comparisons. Exact source and input hashes are checked before the tables are generated.

The source distinction is deliberate. A single publication tree contains both completed studies, while their original numerical execution commits remain visible in the provenance records. Original execution records, arrays, and weights are preserved. New mechanism postprocessing, scalar instrumentation, sensing calculations, and publication checks have their own source identities. The main article, supplement, response, generated tables, and build outputs are therefore reviewable

components of one integration rather than descriptions of an unexecuted future
rewrite.

B2. Resolve the preservation and publication checks. The historical README assertion is now interpreted against its exact historical source. The unchanged historical suites run in an isolated view of their original evidence tree; they do not require a revised root README to masquerade as an earlier document. Nineteen unchanged R12 regression assertions and 104 inherited assertions pass in the recorded replays. Their reports state the source view used, and no historical assertion is weakened or removed.

Current publication preservation is a separate check: every base-tree blob must remain at its original path or at its explicitly declared archive path. The reviewed roots are archived before integration, the original main-root labels are retained, and prior numerical source and evidence files remain unchanged. Compilation is likewise checked on the integrated current documents. The build records each document's PDF hash, page count, unresolved references, duplicate labels, and overfull material, and rejects unresolved defects. Historical test success is not substituted for this current-document check or for an economic result.

B3. Report the direct empirical answer. The completed data answer the comparison at its measured precision. Table III reports direct common-path NBO-minus-DPO and NBO-minus-affine intervals for both initial-state designs; Supplementary Table IV gives every fitted pair. All 120 primary intervals contain zero. Thus the record establishes neither a positive nor a negative direct method difference under the stated simultaneous bounds. Positive raw differences and the 180 positive schedule-relative lower endpoints do not change that conclusion.

This is distinct from the development smoke reviewed by the referee. Each primary final row uses 8,192 independent paths and 1,024 cells. The saved common-state and innovation hashes and schedule arrays are checked before the direct differences are formed. The study retains each fitted stream and its endpoints. Theorem 12 identifies the estimand and its confidence statement; the observed intervals determine what can be concluded about the methods.

B4. Compare work and attained economic precision. Section 12 adds fixed-iteration and fixed-simulator-visit comparisons. NBO, DPO, affine policies, and raw-costate improvement complete eighty iterations; their twentieth and eightieth checkpoints receive full final verification. A twentieth checkpoint uses 81,920 training-state visits and 8,192 validation-state visits; the corresponding counts at eighty iterations are 327,680 and 32,768. The tables also report actor and critic updates because matching simulator visits does not equate derivative work or floating-point operations.

Tables IX and VII compare work to attain common schedule-improvement lower endpoints of zero, 0.0005, and 0.001. Every previously attempted verification is charged. Each full policy-and-schedule verification requires 16,777,216 state transitions. Unreached targets remain in the table. These are common certified-gain targets, rather than observations of the unknown true regret. The paper distinguishes the executed full-fit cost from the retrospective checkpoint-prefix frontier; it does not describe that frontier as a measured early-stopping speedup. It also identifies the operations excluded from recorded clocks.

A second operating-system environment repeats the dimension-ten first-stream cell. The initial-state hashes agree across environments, but the innovation hashes differ. The repeated environment therefore tests execution sensitivity, without being treated as a common-path cross-environment estimator. Within each environment the paired method inputs match exactly. These additions make the work comparison reproducible and expose the limits of a hardware-specific training clock.

B5. Establish and test the critic mechanism. The new welfare inequality in Proposition 11 makes the missing connection explicit. Its performance-difference identity is evaluated under the candidate policy's state law. The bound retains three distinct terms: regression error relative to the conditional rollout mean, the bias between that mean and the continuous-time policy costate, and incomplete feasible-action maximization. Equation 53 gives a separate sufficient costate bound from derivatives of the continuous fixed-policy residual and terminal error, under stated smoothness and global derivative hypotheses.

The empirical mechanism comparison now uses a common independent reference. The saved raw action uses coarse replicate zero. We exclude its paired fine

replicate from the reference mean and evaluate all saved actions against fine replicates one through fifteen. The original arrays and original diagnostics are retained. Tables X and XI show that the critic's MSE exceeds the raw-costate MSE in all six principal panels, with ratios approximately 1.06–1.99; its greedy Hamiltonian gap is also larger throughout. The reference variance and nested-grid sensitivity are reported. These states follow the declared held-out design, so their diagnostics are not inserted into the occupation-weighted welfare theorem. The revision supplies a theorem for the mechanism and an independent test of its fitted implementation; it does not report a critic advantage that the data fail to establish.

B6. Specify a finite-observation implementation. The exact continuous-observation proposition remains an explicitly stated benchmark. Section 10.1 defines the economic and product filtrations, proves that independent private randomization leaves the optimal expected payoff unchanged, and establishes causality of the history reconstruction. Equality of the reconstructed controller's law does not mean pathwise recovery of an unobserved Brownian null component.

Section 10.2 additionally defines a controller receiving a noisy capital observation at a finite set of dates and holding the resulting action between dates. It receives neither Brownian coordinates nor a continuous drift integral. Proposition 15 propagates the sensor moment bound, a global Lipschitz bound for the real reference actor, protected-map arithmetic, clipped innovations, and internal-state error into a payoff allowance in the original continuous economy. Its proof retains cancellation in the accumulated local drift martingale and clipping errors. The eighteen executed cells use two decision grids and three noise levels in each dimension. Their fine-Euler differences diagnose implementation; the separate outward allowance supplies the conditional transfer. A payoff interval transfers only from the exact protected parent on the same grid. The 1,024-cell certificate is not reassigned to a 64- or 128-cell parent.

B7. Delimit and strengthen the absolute reference. The scalar reference has been replayed from its frozen policies on all four original grids and domains. It now reports the original-equation fixed-policy and HJB residuals, Howard checks and iteration counts, boundary values, action-bound frequencies, nested-grid changes,

and losses at five initial states. All twenty-eight original arrays reproduce exactly in the recorded replay environment. Supplementary Section 5.1 derives the finite-grid residual propagation by a discrete maximum principle and states what the observed, non-enclosed residuals measure.

The additional state comparisons matter. On the finest domain-six grid, NBO's loss at zero is approximately 3.44×10^{-6} and DPO's is 1.16×10^{-4} . At initial state one the losses are approximately 1.05×10^{-4} and 5.02×10^{-5} , respectively. The latter ranking favors DPO. The common asymptotic boundary data are not exact unbounded-domain values of each learned policy, and expanding the domain also changes the spacing. These qualifications are part of the reference result. The high-dimensional absolute guarantee remains the independently derived $G_d - L_a$ account in Theorems 7 and 9; no scalar error is extrapolated to high dimension.

B8. State the economic contribution of the whole paper. The revised main text restores the motivating preference-formation problem explicitly. Section 6 states its reflected preference process, consumption and portfolio dynamics, conditional policy rules, cardinal normalization, and valid adjustment-cost envelope. It then explains the recursive-utility, temporal-self, and dynamic-game versions of evaluation and improvement. Their complete derivations and numerical evidence remain in the supplement. The original economic agenda is therefore visible alongside the principal capital verification study.

The capital fee exercise translates a verified gain into a defined resource decision. Equation 58 specifies gross withdrawals, the service's resource charge, and household consumption. The fee threshold is exact for this log-utility model; it is not a claim of empirical calibration or a valuation of processor time in dollars. Its role is to make the verified utility difference interpretable. The paper's methodological contribution combines a specified Bellman computation, economic error bounds, an observation transfer, and reproducible method comparisons. These claims are evaluated on their mathematics and measured results, without treating a broad ranking of algorithms or an unobserved empirical economy as an established conclusion.

MAJOR COMMENTS

M1. Identify the authoritative submitted paper. The root `ECTA.tex` and `supp.tex` are the authoritative integrated sources. The response and generated R14 tables use their section, proposition, theorem, and table labels. The revision guide identifies the compiled outputs and the raw-record locations. The integration script is retained as a reproducibility tool; it is not the only representation of the article. Its editorial map records archived roots, preserved labels, and intact relocations.

M2. Separate protocols, execution, and new calculations. The report's distinction is implemented throughout the revision. The original primary and fixed-work protocols identify their own generating sources and worker runs. The response reports numerical outcomes only from the complete records and their independent replay. The present version adds mathematical results, corrected postprocessing, deterministic reference replay, and a sensing diagnostic; it reports no new training. Development smoke remains a development artifact. The two numerical generating commits and the publication integration are recorded separately, including the scope of each test report.

M3. Provide every direct contrast before aggregation. Supplementary Tables I and IV report all primary policy and direct-comparison rows. Each direct row includes the paired mean and continuous-time lower and upper endpoints. The main summaries identify the numbers of positive, negative, and inconclusive fitted comparisons. Work is supplied by the associated cost and seed records. All 120 primary direct intervals are inconclusive. Across-seed summaries describe ten specified fitted objects; the paper does not posit an optimizer-population probability model to turn their frequency into another confidence statement.

M4. Add fixed-work and common-target comparisons. The completed extension supplies both twenty- and eighty-iteration checkpoints, exact simulator visits, optimizer update counts, work-to-target tables, and a second-environment comparison; see Tables VIII, VII, and VIII. At target 0.0005, four of six NBO streams, two DPO streams, no affine streams, and four raw-costate streams attain the target among the declared checkpoints. No method attains 0.001 in these six cells. These are

descriptive endpoint-attainment counts. They do not override the direct payoff intervals or imply that an unreached target is unattainable with another budget.

Verification includes the policy and schedule trajectories and all earlier attempted checks. Recorded elapsed components and exact transition counts are reported separately. This provides a common economic target and a reproducible work account, while preserving the difference between simulator work, optimizer work, and complete execution time.

M5. Connect the mechanism quantities. Section 9.2 identifies the required mathematical chain: true continuation derivatives, feasible Hamiltonian improvement, and the candidate's economic payoff. The independent panels measure rollout variation, corrected costate MSE, and action loss against a common fine-grid reference; the fixed-work tables give the independently evaluated policy payoffs and computational work. The critic's larger errors in every principal panel are explicit. The raw-costate action diagnostic for a frozen NBO policy is distinguished from the separately trained raw-costate comparator. Neither reference noise nor a finite nested-grid difference is treated as a certified continuous-time costate error.

M6. Define the comparator filtration. Proposition 14 works on a canonical product space with economic Brownian motion and independent private randomization. Conditioning on a fixed randomizer realization leaves an admissible nonanticipative Brownian control, so integrating conditional payoffs cannot increase the supremum. The reverse inclusion follows by ignoring the randomizer. This establishes equality of the stated suprema without requiring an optimizer or a maximizing realization of the randomizer. The anchor proof also applies directly in the enlarged filtration. The information assumptions and observation clock are specified before the implementation results.

M7. Distinguish exact and finite observations. The continuous-history reconstruction and the finite-measurement policy are now different explicitly defined controllers. The latter uses noisy observations at decision dates, protected saved weights, and held actions. Its continuous-economy allowance is proved in Proposition 15 and Supplementary Section 3. The eighteen numerical cells retain the sensor

law, paths, decision mesh, fine-Euler mesh, hashes, and arithmetic account. Their simulated differences do not create new statistical payoff endpoints or certify a different parent controller.

M8. Report the full scalar diagnostics. The four-grid replay records every fixed-policy solve and Howard step. The supplementary reference account gives five-state losses, complete common-domain changes and their locations, optimal-policy changes, endpoint boundary residuals, and action-bound frequencies. The realized action-bound frequency is zero for all reported scalar policies, and each time step uses three Howard iterations in the replay. The finite-grid equation residual propagation is stated against the exact solution of the same discrete problem. Its diagnostic substitution uses observed floating-point residuals, while boundary approximation and continuum consistency remain separate terms.

M9. Clarify the contribution through organization. The main article now follows the shared model and NBO map, the general verification results, the original economic applications, the capital benchmark, the costate and observation results, and the executed evidence. The compact application section explains why the preference, recursive, temporal, and game problems belong to this framework. The supplement has a reading guide and complete proof and numerical records. Earlier learning specifications and experiments are identified as earlier records, so they are not mistaken for the current primary design.

This organization addresses the difficulty of locating the theorem–algorithm–experiment chain while preserving the original topic and substantive applications. It does not remove unsuccessful policies, redefine a historical experiment after observing its results, or use the capital theorem to supply hypotheses missing from another application. The editorial map and exact archived roots make the reorganization auditable.

M10. Retain the evidentiary discipline. The revision retains development/final noise distinctions, all declared fitted streams, raw arrays, weights, checkpoints, adverse outcomes, state stresses, input hashes, and common-path cancellation checks. The primary and fixed-work studies use 728 and 196 one-sided statements,

respectively, for a combined 924 of the 4,000 allocated statements at the declared confidence level. The new mechanism, scalar-reference, and sensing diagnostics add no stochastic confidence endpoints. The cross-environment innovation mismatch is disclosed rather than overwritten. Numerical completion, reproduction of an array, a positive schedule comparison, and a signed direct-method comparison are separately recorded facts.

CLOSING RESPONSE

The revised NBO paper gives the referee a complete empirical record and a stronger mathematical connection between policy evaluation, feasible improvement, observation, and economic performance. Its positive schedule-improvement results, unresolved direct rankings, and unfavorable critic diagnostics are all part of that record. The original economic models remain in the article's scope and full supplementary development. The claims advanced for further review are the specified Bellman method, its model-appropriate verification results, the new welfare and finite-observation accounts, and the executed economic comparisons.